Run report: ecbench session `ECBS1h4687f1d786c2` (k4-p29-d4cost), 2026-10-14T20:07Z

Spec `specs/SPEC-k4-p29-d4cost.json` (id `ECS1h9deb3f4c9890`). Command `ecbench run --cpus auto --wait --allow-busy`; binary sha256 `67b8cc95…3e22`, commit `2b49f38d1`. Host: Xeon @ 2.10 GHz, 4 vCPU, one reserved core. The ladder ran beside the session on the other cores.

Workloads: `fp4_torsion(29, 1, 16)`, `icv1-fp5k4-tm926-7bc5b359`, $r = 44263$, targets `Wc87e324798ff` and `W1016932647b2` (seed 202610134029). One round, alternate order. 8 executions requested, 8 completed, 8 verified.

arm	method	target 0 S	target 1 S	wall (s)	level
rho-neg	rho.negation	1.065	1.412	0.01	L2
ic-d4-committed	`gaudry-d4:macaulay=14,charge_unsolved=1`	657,144,598	722,837,732	7,321 / 7,928	L1
ic-d4-chains	… `macaulay=26,weighted=1,chains=1,sparsest=1,replay=1,charge_unsolved=1`	1,653,062	1,779,983	221 / 235	L1
rho-neg-aa	rho.negation	1.065	1.412	0.01	L2

Recorded counters for the IC arms on targets 0 / 1: 574 / 635 trials in both arms; 9 / 8 relations; unsolved systems 9 / 8 (committed) and 11 / 13 (new); shape replays 568 / 625 with 3 / 4 fallbacks; solver ops $1.341 \cdot 10^{13}$ / $1.475 \cdot 10^{13}$ (committed) and $3.373 \cdot 10^{10}$ / $3.632 \cdot 10^{10}$ (new) $\mathbb{F}_p$ multiplications, priced at 97 per addition.

`ecbench compare` (`comparisons/`): ic-d4-chains / ic-d4-committed = 0.0025 in ops (cluster interval over 2 targets); ic-d4-chains / rho-neg = $1.39 \cdot 10^6$ (1.26–1.55). No wall comparison was admitted: no pair reached L2. Audit: `audits/k4-p29-d4cost.audit.json` (`verify --replay-all`).

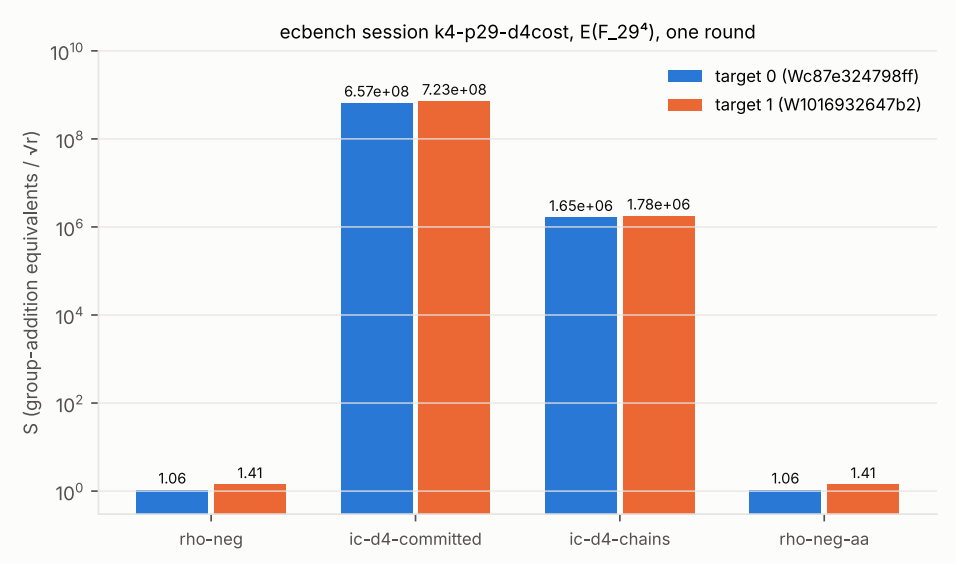


Figure 1: S per arm and target as recorded.